

Kia/Hyundai Auto Theft Surge: Manufacturer Risk & Response

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Agenda

- Audience & Objective
- Data & Methods
- Key Findings
- Implications for Kia/Hyundai
- Recommended Actions
- Ethical Considerations
- Code Appendix

Audience: Kia & Hyundai Corporate Leadership

Who This Briefing is For

- Product Safety Leadership
- Risk Management Teams
- Corporate Legal & Compliance
- Engineering & Security Teams
- Brand Reputation & Communications

What They Need to Know

- Clear, actionable insights
- Evidence-based risk framing
- City-level exposure patterns
- Brand-specific vulnerability
- Recommendations tied to mitigation

Objective & Call to Action

Objective

- Quantify the Kia/Hyundai theft surge
- Compare theft patterns to other makes
- Identify high-risk cities
- Highlight brand-specific vulnerability
- Provide data-driven insight for decision-making

Call to Action

Manufacturers should implement a coordinated national response:

- Security upgrades (software + hardware)
- Owner outreach and communication
- Dealer-level mitigation strategies
- City-specific interventions
- Proactive brand and legal risk management

Data & Methods

Four Primary Datasets Used

- `carTheftsMap.csv` - National auto theft change (2019-2022)
- `KiaHyundaiMilwaukeeData.csv` - Milwaukee long-term theft trends
- `kiaHyundaiThefts.csv` – City-level Kia/Hyundai vs Other theft counts
- **Motherboard VICE News Kia Hyundai Theft Data.xlsx** – National city-level theft dataset

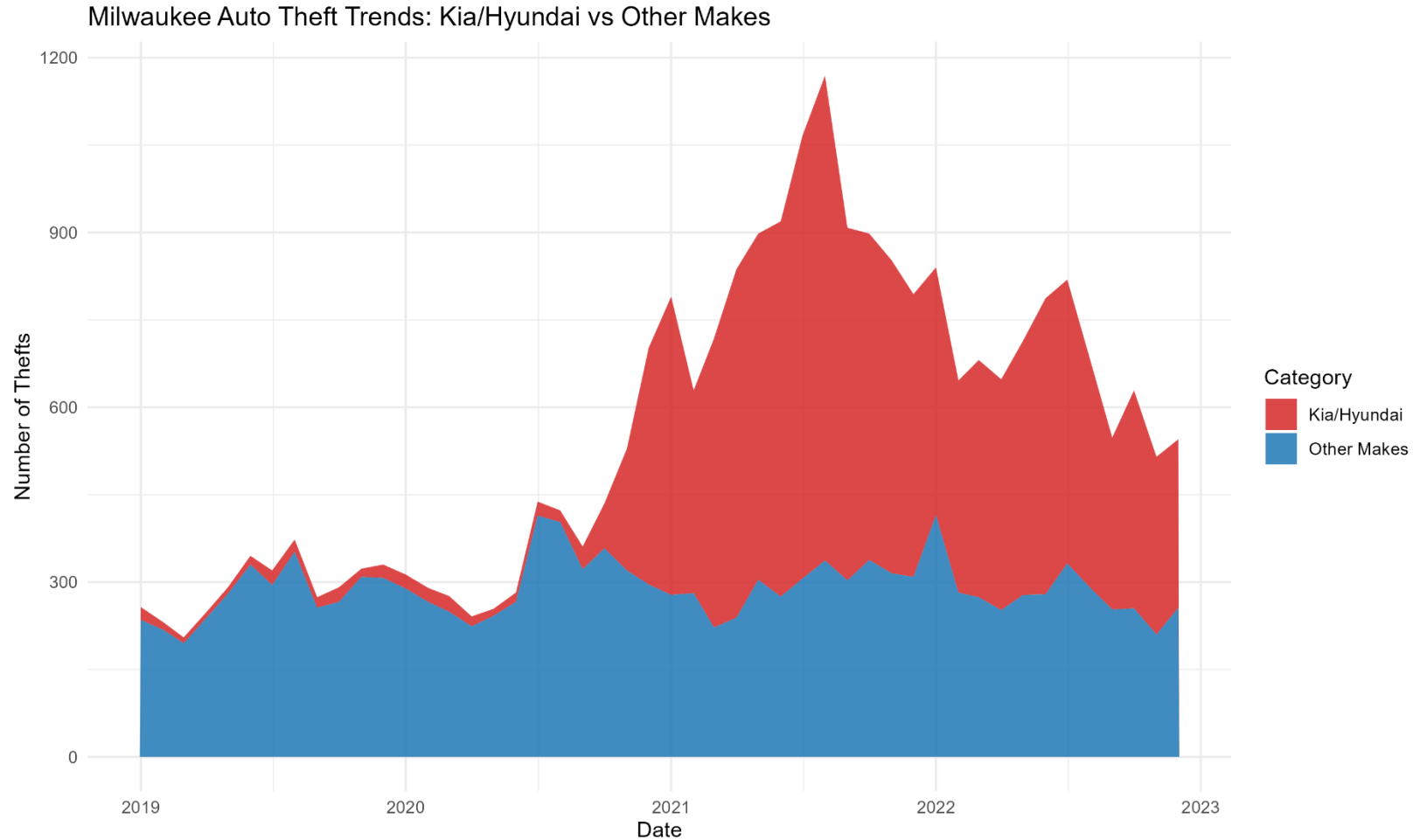
Key Transformations

- Reconstructed multi-row headers into usable column names
- Converted all numeric fields
- Created “Other” thefts = All – Kia/Hyundais
- Filtered to the latest full year for fair cross-city comparison
- Aggregated Kia/Hyundai totals for treemap and stacked bar

Tools & Workflow

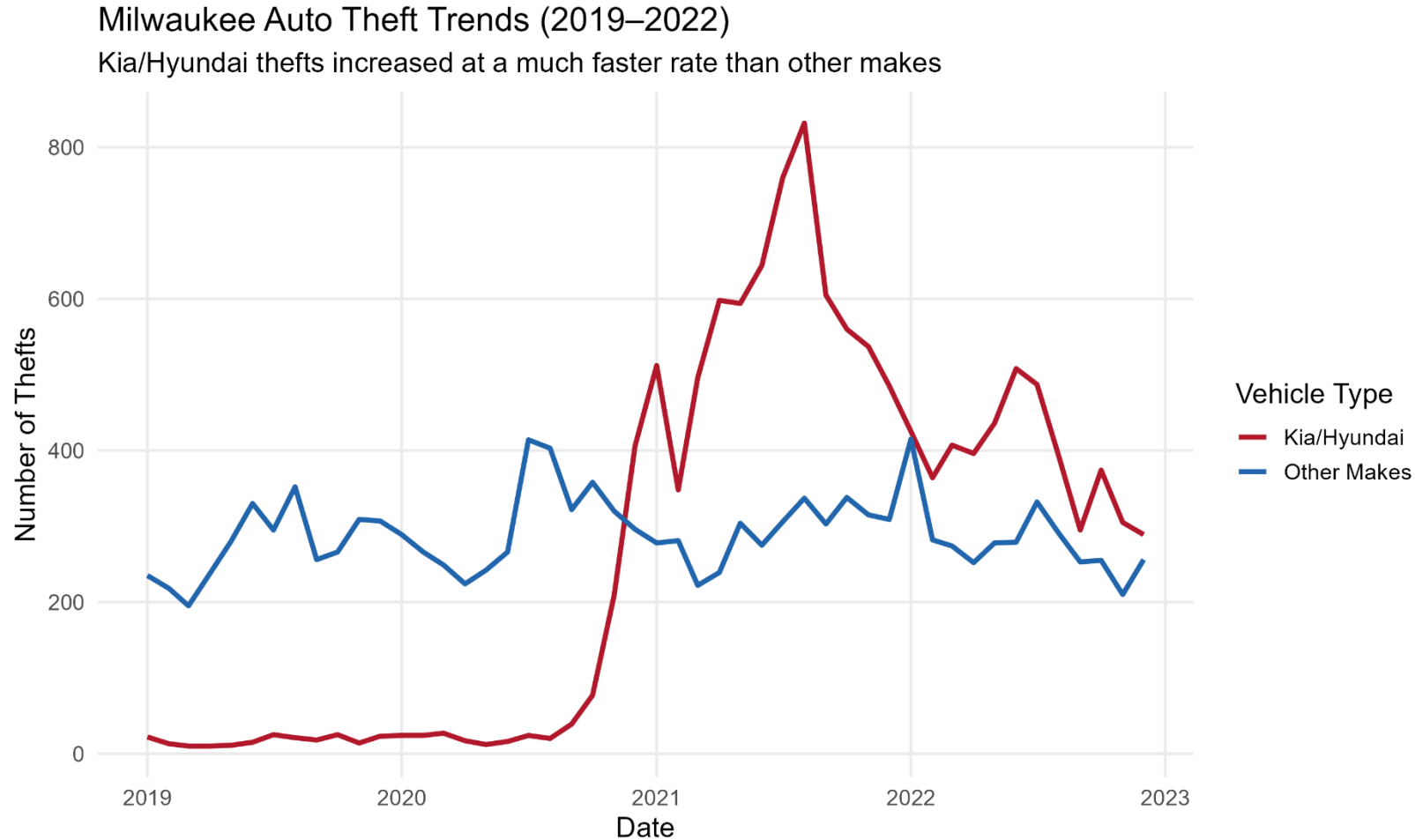
- R (tidyverse, ggplot2, lubridate)
- Seven modular scripts for cleaning, transforming, and visualizing
- All scripts included in the Code Appendix

Milwaukee Theft Trends (2019-2023)



Kia/Hyundai thefts surged sharply beginning in 2021, while other makes remained relatively stable.

Timing & Magnitude of the Surge

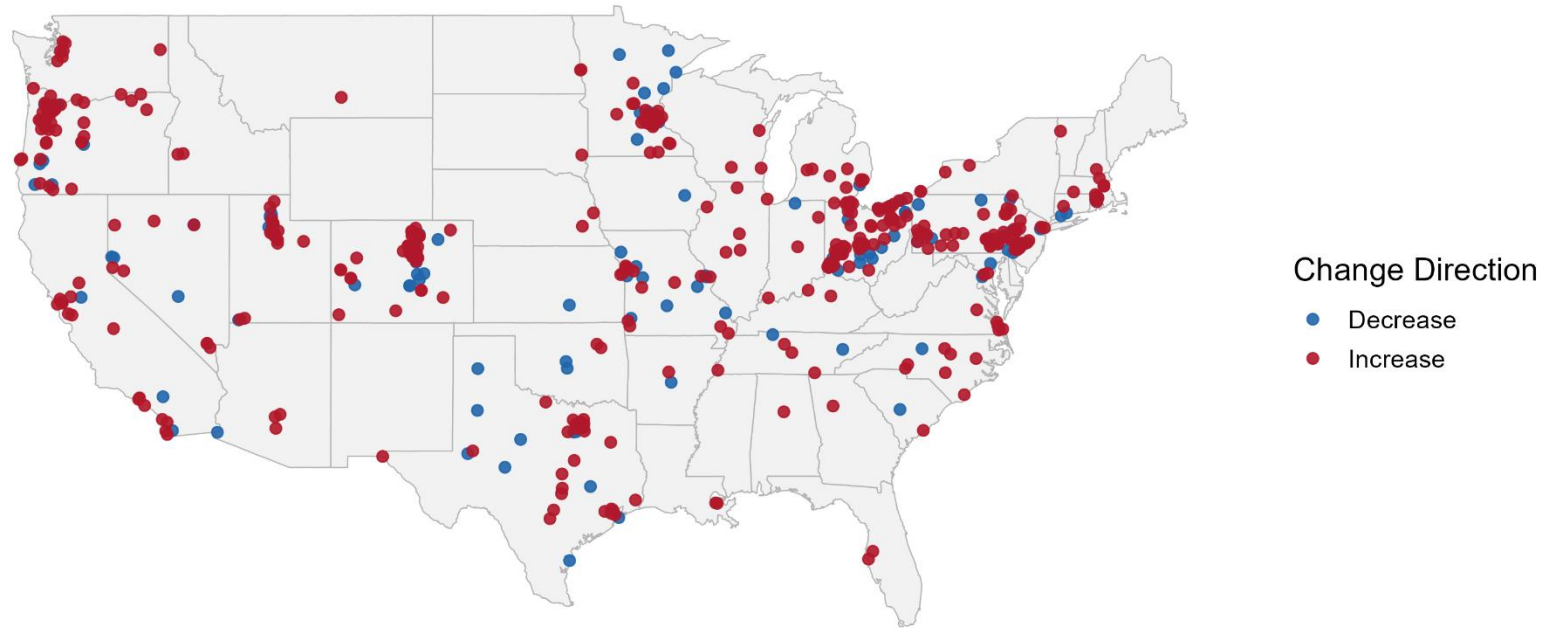


The spike in thefts is steep, recent, and isolated to Kia/Hyundai vehicles.

National Auto Theft Change (2019-2022)

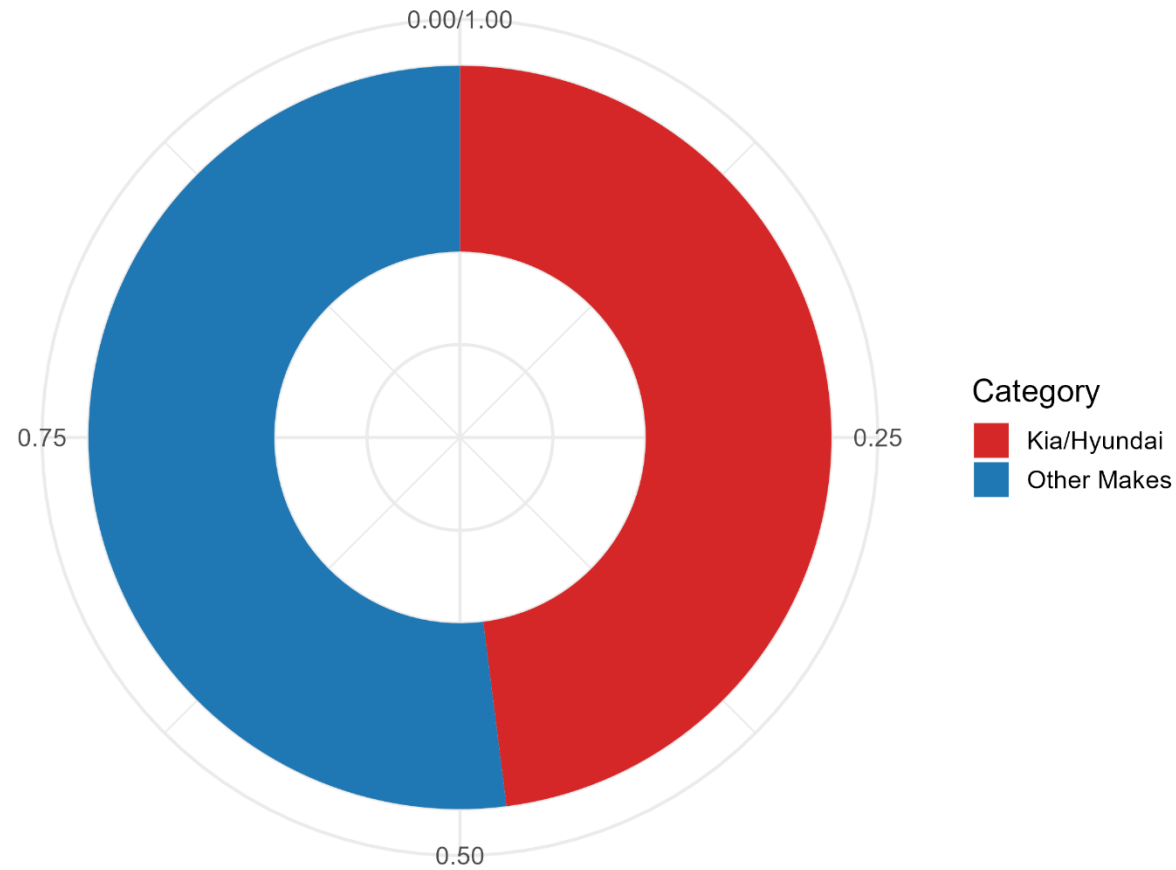
Direction of Auto Theft Change (2019–2022)

Red = increase, Blue = decrease



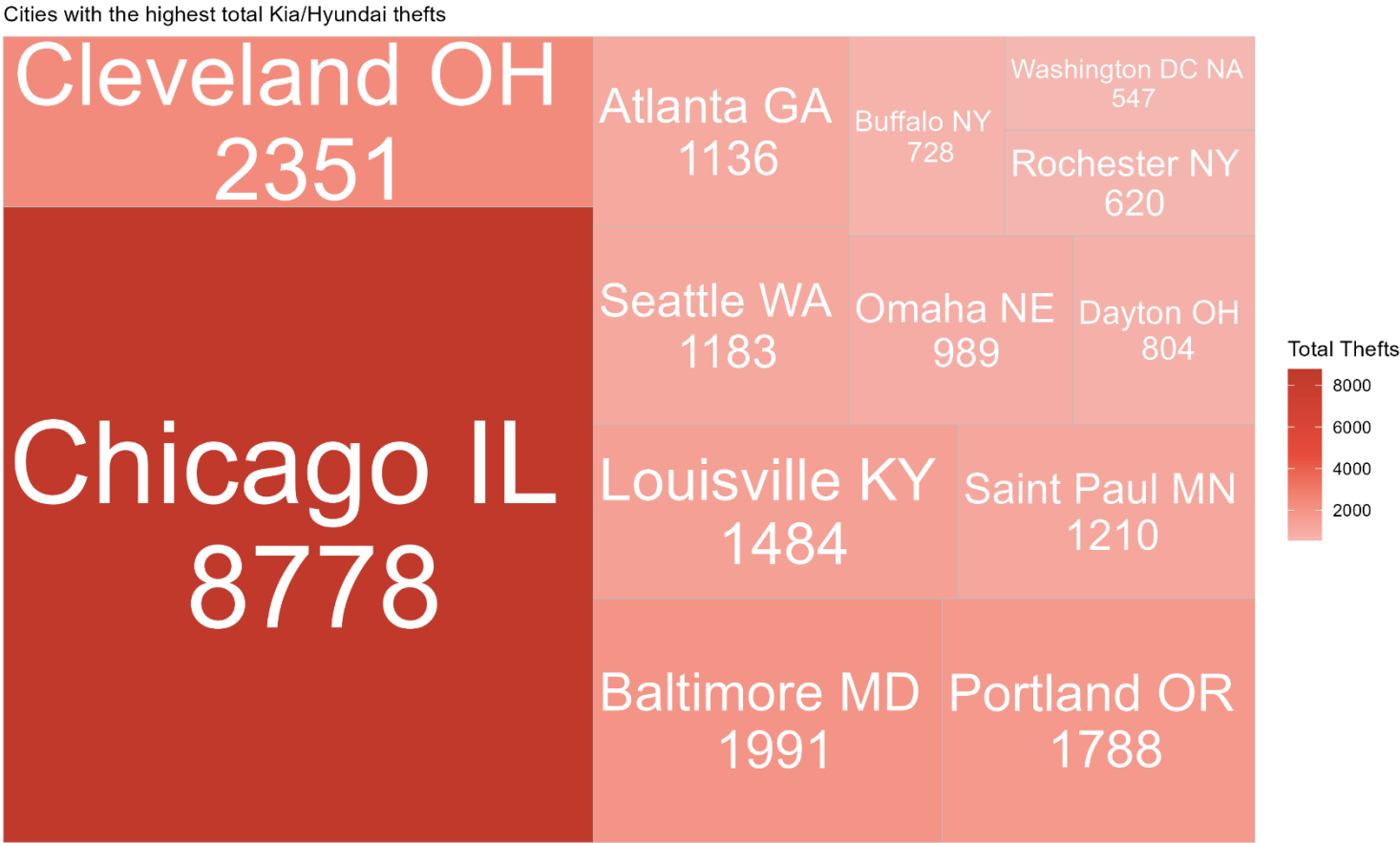
National auto theft rose significantly from 2019–2022, creating conditions that amplified brand-specific vulnerabilities.

Share of Kia/Hyundai Thefts 2022



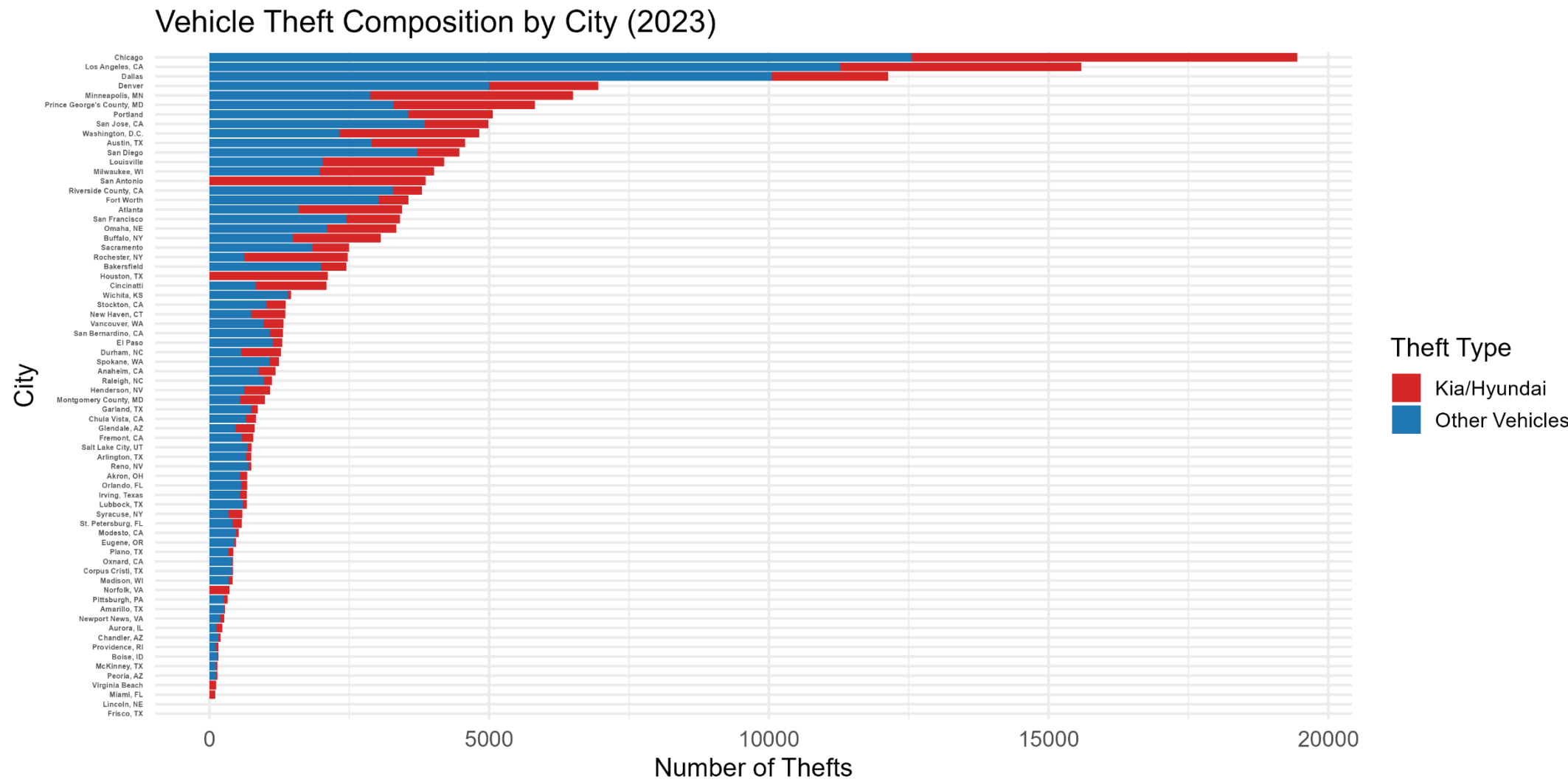
Kia/Hyundai vehicles accounted for a substantial share of total thefts in 2022, far exceeding their market presence.

City-Level Concentration of Kia/Hyundai Thefts



A small number of cities account for a disproportionately large share of Kia/Hyundai thefts.

Kia/Hyundai vs Other Makes Across Cities



In several cities, Kia/Hyundai thefts make up a large share of total thefts, highlighting concentrated brand-specific risk.

Interpretation of Findings

The surge is recent, steep, and make specific.

- Kia/Hyundai thefts accelerated sharply beginning in 2021, when other makes remained relatively stable across the same period.

The issue is concentrated in specific cities.

- A small number of urban areas account for a disproportionate share of Kia/Hyundai thefts. This indicates localized hotspots rather than an overall national trend.

National auto theft increases amplified the problem.

- Nationwide theft growth from 2019–2022 increased the visibility of make-specific patterns in the data.

These patterns point to make-focused solutions and not general approaches.

Recommendations

Prioritize make-focused mitigation efforts.

- Support actions directed at Kia/Hyundai vehicles, including software updates, owner communication, and dealer-level assistance.

Concentrate resources in high-impact cities.

- Direct prevention, outreach, and coordination efforts toward the urban areas driving most of the theft activity.

Strengthen monitoring of emerging theft patterns.

- Continue tracking theft trends to identify shifts early and to adjust responses as needed.

These steps align resources with where the issue is most concentrated and most actionable.

Narrative

Narrative Summary

In my analysis there was a make-specific rise in Kia and Hyundai thefts starting in 2021. The other makes stay mostly steady over the same timeframe and that makes the change for Kia and Hyundai pretty obvious. The increase is pretty focused to a few larger cities. It isn't spread across the country, so it doesn't appear to be a national trend. When compared with the overall growth of auto thefts in U.S., the greater increase for these two makes is much larger.

Design Approach

My approach to building the visuals took into account many of the Gestalt principles. They will help the viewer understand the story rather quickly and not require digging into data. I used similarity in color representation to keep Kia/Hyundai consistent. This way the viewer doesn't have to relearn the color patterns for each visualization. The figure-ground principle helped me keep the main data the focus. I used proximity and alignment keep related elements together. This helps the viewer move from visual to visual and idea to idea without losing their place in the story.

Ethical Considerations

Throughout the project, I stayed focused on what the data actually supports. I'm describing the patterns in the data. I wasn't looking for causes or behaviors. The data is aggregated at the city level, so no individuals or specific locations are known. I avoided language that could direct the findings in a specific direction or imply blame. My interpretations were tied directly to visible trends. My intent was to inform, not shape the discussion in ways the data can't support.

Appendix: R Code for Reproducibility

All scripts used to import, clean, analyze, and visualize the data.

All code is shown exactly as executed. Each script is included on its own slide for clarity and reproducibility.

Code: Data Import & Cleaning

```
1 library(tidyverse)
2 library(lubridate)
3 library(ggplot2)
4 library(treemapify)
5 library(sf)
6 library(readxl)
7 library(dplyr)
8
9 # Load datasets
10 kia_mke <- read_csv("data/KiaHyundaiMilwaukeeData.csv")
11 car_map <- read_csv("data/carTheftsMap.csv")
12 kia_thefts <- read_csv("data/KiaHyundaiThefts.csv")
13 vice_data <- read_excel("data/Motherboard VICE News Kia Hyundai Theft Data.xlsx")
14
15 # Quick preview
16 glimpse(kia_mke)
17 glimpse(car_map)
18 glimpse(kia_thefts)
19 glimpse(vice_data)
20
21 # kia_mke data feature engineering
22 # Create date column for time series analysis
23 kia_mke <- kia_mke %>%
24   mutate(
25     month_num = match(str_to_title(toupper(month)), month.abb),
26     date = make_date(year, month_num, 1)
27   )
28
29
30
31
32 |
```


Code: Auto Theft Stacked Area Chart

```
1 library(tidyverse)
2 library(lubridate)
3 library(ggplot2)
4
5 # Create long format for stacked area chart
6 kia_mke_long <- kia_mke %>%
7   select(date, countKiaHyundaiThefts, countOtherThefts) %>%
8   mutate(
9     countKiaHyundaiThefts = as.numeric(countKiaHyundaiThefts),
10    countOtherThefts = as.numeric(countOtherThefts)
11  ) %>%
12  pivot_longer(
13    cols = c(countKiaHyundaiThefts, countOtherThefts),
14    names_to = "category",
15    values_to = "thefts"
16  )
17
18 # Plot
19 print(
20   ggplot(kia_mke_long, aes(x = date, y = thefts, fill = category)) +
21     geom_area(alpha = 0.85) +
22     labs(
23       x = "Date",
24       y = "Number of Thefts",
25       fill = "Category"
26     ) +
27     scale_fill_manual(
28       values = c(
29         "countKiaHyundaiThefts" = "#d62728",
30         "countOtherThefts" = "#1f77b4"
31       ),
32       labels = c(
33         "countKiaHyundaiThefts" = "Kia/Hyundai",
34         "countOtherThefts" = "Other Makes"
35       )
36     ) +
37     theme_minimal()
38 )
39
40 ggsave(
41   filename = "plots/stacked_area_kia_mke.png",
42   plot = last_plot(),
43   width = 10,
44   height = 6,
45   dpi = 300
46 )
47
48
```

Code: Auto Theft Trend Lines

```
1 library(tidyverse)
2 library(ggplot2)
3 library(lubridate)
4
5 # Prep data for plotting
6 kia_mke_long <- kia_mke %>%
7   select(date, countKiaHyundaiThefts, countOtherThefts) %>%
8   pivot_longer(
9     cols = c(countKiaHyundaiThefts, countOtherThefts),
10    names_to = "type",
11    values_to = "count"
12  ) %>%
13  mutate(
14    type = recode(type,
15                  "countKiaHyundaiThefts" = "Kia/Hyundai",
16                  "countOtherThefts" = "Other Makes")
17  )
18
19 # Build line chart
20 p <- ggplot(kia_mke_long, aes(x = date, y = count, color = type)) +
21   geom_line(linewidth = 1.2) +
22   scale_color_manual(
23     values = c(
24       "Kia/Hyundai" = "#B2182B",
25       "Other Makes" = "#2166AC"
26     )
27   ) +
28   labs(
29     title = "Milwaukee Auto Theft Trends (2019-2022)",
30     subtitle = "Kia/Hyundai thefts increased at a much faster rate than other makes",
31     x = "Date",
32     y = "Number of Thefts",
33     color = "Vehicle Type"
34   ) +
35   theme_minimal(base_size = 14) +
36   theme(
37     panel.grid.minor = element_blank(),
38     plot.background = element_rect(fill = "white", color = NA)
39   )
40
41 print(p)
42
43 # Save plot
44 ggsave("plots/milwaukee_kia_hyundai_trend.png", p, width = 10, height = 6, dpi = 300)
45
```

Code: Auto Theft Direction Map

```
1 library(tidyverse)
2 library(ggplot2)
3 library(sf)
4 library(maps)
5
6 # Compute direction of change
7 car_map <- car_map %>%
8   mutate(
9     abs_change = countCarThefts2022 - countCarThefts2019,
10    direction = if_else(abs_change > 0, "Increase", "Decrease")
11  )
12
13 # Remove Alaska
14 car_map <- car_map %>%
15   filter(latitude < 50)
16
17 # Convert to sf
18 car_map_sf <- st_as_sf(
19   car_map,
20   coords = c("longitude", "latitude"),|
21   crs = 4326
22 )
23
24 # Load US basemap
25 us_map <- st_as_sf(map("state", plot = FALSE, fill = TRUE))
26
27 # Build map
28 p <- ggplot() +
29   geom_sf(data = us_map, fill = "gray95", color = "gray70") +
30   geom_sf(
31     data = car_map_sf,
32     aes(color = direction),
33     size = 2,
34     alpha = 0.9
35   ) +
36   scale_color_manual(
37     values = c(
38       "Increase" = "#B2182B",
39       "Decrease" = "#2166AC"
40     )
41   ) +
42   labs(
43     title = "Direction of Auto Theft Change (2019-2022)",
44     subtitle = "Red = increase, Blue = decrease",
45     color = "Change Direction"
46   ) +
47   theme_minimal(base_size = 14) +
48   theme(
49     panel.background = element_rect(fill = "white", color = NA),
50     plot.background = element_rect(fill = "white", color = NA),
51     panel.grid = element_blank(),
52     axis.title = element_blank(),
53     axis.text = element_blank(),
54     axis.ticks = element_blank()
55   )
56
57 print(p)
58
59 # Save plot
60 ggsave("plots/map_direction_change.png", p, width = 10, height = 6, dpi = 300)
61
```

Code: Auto Theft Share Donut Chart

```
1 library(tidyverse)
2 library(ggplot2)
3
4 # Summarize totals
5 make_share <- kia_mke %>%
6   summarise(
7     kia_hyundai = sum(countKiaHyundaiThefts, na.rm = TRUE),
8     other = sum(countOtherThefts, na.rm = TRUE)
9   ) %>%
10  pivot_longer(everything(), names_to = "category", values_to = "count") %>%
11  mutate(
12    fraction = count / sum(count),
13    ymax = cumsum(fraction),
14    ymin = lag(ymax, default = 0)
15  )
16
17 # Plot
18 p <- ggplot(make_share, aes(ymax = ymax, ymin = ymin, xmax = 4, xmin = 3, fill = category)) +
19   geom_rect() +
20   coord_polar(theta = "y") +
21   xlim(c(2, 4)) +
22   theme_minimal(base_size = 14) +
23   theme(
24     panel.background = element_rect(fill = "white", color = NA),
25     plot.background = element_rect(fill = "white", color = NA),
26     axis.text.y = element_text(color = "white"),
27     axis.title.y = element_text(color = "white")
28   ) +
29   scale_fill_manual(
30     values = c(
31       "kia_hyundai" = "#d62728",
32       "other" = "#1f77b4"
33     ),
34     labels = c(
35       "kia_hyundai" = "Kia/Hyundai",
36       "other" = "other Makes"
37     )
38   ) +
39   labs(
40     fill = "category"
41   )
42
43
44 print(p)
45
46 ggsave("plots/donut_kia_mke.png", p, width = 8, height = 6, dpi = 300)
47
```

Code: Auto Theft Treemap

```
1 library(tidyverse)
2 library(treemapify)
3
4 # Summarize Kia/Hyundai thefts by city
5 kia_tree <- kia_thefts %>%
6   group_by(city, state) %>%
7   summarise(total_kia_hyundai = sum(countKiaHyundaiThefts, na.rm = TRUE)) %>%
8   ungroup()
9
10 # Plot treemap with red gradient
11 p5 <- ggplot(kia_tree, aes(
12   area = total_kia_hyundai,
13   fill = total_kia_hyundai,
14   label = paste(city, state, "\n", total_kia_hyundai)
15 )) +
16   geom_treemap() +
17   geom_treemap_text(colour = "white", place = "centre", grow = TRUE) +
18   scale_fill_gradientn(
19     colors = c("#F5B7B1", "#E74C3C", "#C0392B"),
20     name = "Total Thefts"
21   ) +
22   labs(
23     subtitle = "Cities with the highest total Kia/Hyundai thefts"
24   ) +
25   theme_minimal()
26
27 print(p5)
28
29 ggsave("plots/treemap_kia_thefts.png", p5, width = 10, height = 6, dpi = 300)
30
```

Code: Auto Theft Stacked Bar Chart

```
1 library(tidyverse)
2 library(readxl)
3 library(ggplot2)
4 library(lubridate)
5
6 # Load VICE dataset
7 vice_raw <- read_excel("data/Motherboard VICE News Kia Hyundai Theft Data.xlsx")
8
9 # Extract subheaders
10 subheaders <- as.character(vice_raw[1, ])
11
12 # Build new headers
13 original_headers <- colnames(vice_raw)
14 new_headers <- original_headers
15
16 for (i in seq(2, length(original_headers), by = 3)) {
17   city <- original_headers[i]
18   new_headers[i] <- paste0(city, "_", subheaders[i])
19   new_headers[i+1] <- paste0(city, "_", subheaders[i+1])
20   new_headers[i+2] <- paste0(city, "_", subheaders[i+2])
21 }
22
23 colnames(vice_raw) <- new_headers
24
25 # Rename first column
26 colnames(vice_raw)[1] <- "Date"
27
28 # Remove subheader row
29 vice_clean <- vice_raw[-1, ]
30
31 # Convert Date
32 vice_clean$date <- as.Date(vice_clean$date)
33
34 # Convert other columns to numeric
35 vice_numeric <- vice_clean %>%
36   mutate(across(-date, ~ suppressWarnings(as.numeric(.))))
37
38 # Get latest year
39 latest_year <- max(year(vice_numeric$date), na.rm = TRUE)
40
41 vice_latest <- vice_numeric %>%
42   filter(year(date) == latest_year)
43
44 # Identify cities
45 kia_cols <- grep("_Kia/Hyundai$", names(vice_latest), value = TRUE)
46 cities <- sub("_Kia/Hyundai$", "", kia_cols)
47
48 # Build Kia + Other dataset
49 vice_stack <- map_dfr(cities, function(city) {
50   kia_col <- paste0(city, "_Kia/Hyundai")
51   all_col <- paste0(city, "_All")
52   tibble(
53     city = city,
54     kia = vice_latest[[kia_col]],
55     other = vice_latest[[all_col]] - vice_latest[[kia_col]]
56   )
57 })
58
```

```
58
59 # Collapse to totals
60 vice_stack_totals <- vice_stack %>%
61   group_by(city) %>%
62   summarise(
63     kia = sum(kia, na.rm = TRUE),
64     other = sum(other, na.rm = TRUE),
65     total = kia + other
66   )
67
68 # Reshape for stacked bar
69 vice_stack_long <- vice_stack_totals %>%
70   pivot_longer(
71     cols = c(kia, other),
72     names_to = "type",
73     values_to = "count"
74   )
75
76 # Build stacked bar chart
77 p6 <- ggplot(
78   vice_stack_long,
79   aes(x = reorder(city, total), y = count, fill = type)
80 ) +
81   geom_bar(stat = "identity") +
82   coord_flip() +
83   labs(
84     title = paste0("Vehicle Theft Composition by City (", latest_year, ")"),
85     x = "City",
86     y = "Number of Thefts",
87     fill = "Theft Type"
88   ) +
89   scale_fill_manual(
90     values = c("kia" = "#d62728", "other" = "#1f77b4"),
91     labels = c("Kia/Hyundai", "Other Vehicles")
92   ) +
93   theme_minimal(base_size = 14) +
94   theme(
95     axis.text.y = element_text(size = 4, face = "bold"),
96     axis.text.x = element_text(size = 12),
97     legend.position = "right"
98   )
99
100 print(p6)
101
102 # Save plot
103 ggsave("plots/stacked_bar_vice_latest_year.png", p6, width = 12, height = 6, dpi = 300)
104
```

40:1 (Top Level) ↕

R Script